EnterpriseOS

Business Requirements Document (BRD) + Software Requirements Specification (SRS)

Phase 2: Core Workforce and Project Operations

Document control	Value
Version	1.0 - Client review draft
Status	Proposed; client confirmation required
Project	EnterpriseOS - multi-tenant SaaS ERP
Prepared for	Client / Business Sponsor
Stack	React + TypeScript; Java Spring Boot; MySQL 8.4 LTS
Prerequisite	Approved and delivered Phase 1 baseline
Date	11 November 2026
Approval	Pending client sign-off

Purpose: Determine client decisions, specify the proposed phase scope and provide verifiable approval and acceptance gates. Any default described here is a proposal, not a confirmed client requirement.

1. Executive Summary & Business Outcomes

Deliver employee directory and lifecycle, organizational hierarchy, shifts, attendance, leave management and project/task execution on top of Phase 1 tenant and RBAC foundation.

Phase 2 consumes the shared identity, tenancy and role model defined in Phase 1. Functional changes are implemented as new bounded modules in the same Spring Boot modular monolith. Existing security controls and audit records remain mandatory.

2. Prerequisites, Dependencies & Stakeholders

Dependency	Entry requirement
Prior phase	Phase 1 released, accepted and regression-tested.
Tenant & identity	Active tenant context, membership validation, session security and permission model available.
Reference data	Departments, staff, currencies, policies and other dependent master data approved as applicable.
Stakeholders	Client sponsor, product owner, department SMEs, information security contact, UAT owner nominated.
Delivery assets	Source repository, staging environment, MySQL migration process, CI tests, email service available.

Responsible stakeholders: HR Admin, HR Officer, Department Manager, Project Manager, Team Member, Employee, Tenant Auditor.

3. Client Discovery - Required Inputs Before Scope Freeze

1. Which employee fields are mandatory and which are legally sensitive?
   Client answer / reference: _____

2. How many branches, departments, employment types and approval levels exist?
   Client answer / reference: _____

3. How is time recorded: manual, import or biometric integration (integration excluded by default)?
   Client answer / reference: _____

4. What are leave accrual, carry-forward, holidays and weekend calculation policies?
   Client answer / reference: _____

5. Are overtime calculations informational or payable? (Payroll deferred to Phase 3.)
   Client answer / reference: _____

6. Does the client need sprint methodology, timesheets, or just Kanban tasks?
   Client answer / reference: _____

7. Who can see other teams, salaries or private HR documents?
   Client answer / reference: _____

8. What is the historical employee/attendance import volume and source format?
   Client answer / reference: _____

4. Functional Requirements - Traceability Baseline

4.1 Employee & Department Management

- P2-FR-0101 | Create and manage branches, departments, teams, reporting hierarchy and job designations
- P2-FR-0102 | Create employee profiles with employee code, employment status, joining date and manager
- P2-FR-0103 | Onboard, transfer, change position and offboard employees with effective dates
- P2-FR-0104 | Provide employee self-service for personal profile and emergency contacts
- P2-FR-0105 | Control visibility of sensitive employee fields using field-level permissions

4.2 Attendance & Shift Scheduling

- P2-FR-0201 | Configure locations, workdays, holidays and shift templates
- P2-FR-0202 | Assign shifts and maintain effective-dated rosters
- P2-FR-0203 | Capture manual check-in/out with source, timestamps and editor tracking
- P2-FR-0204 | Compute daily presence, lateness, overtime and exceptions by approved rules
- P2-FR-0205 | Allow attendance correction requests with supervisor review

4.3 Leave & Holiday Management

- P2-FR-0301 | Configure leave types, entitlement, carry-forward and accrual policy
- P2-FR-0302 | Publish balances and year-specific leave rules
- P2-FR-0303 | Submit, cancel and track leave requests with overlapping-date validation
- P2-FR-0304 | Route manager and HR approvals with delegation and escalation
- P2-FR-0305 | Update balances atomically and retain adjustments history

4.4 Project, Sprint & Task Management

- P2-FR-0401 | Create projects, teams and assignments with scoped permissions
- P2-FR-0402 | Manage milestones, backlog, sprint and Kanban boards
- P2-FR-0403 | Manage task priorities, labels, assignees, dependencies and comments
- P2-FR-0404 | Capture task activity and optional manual time entries
- P2-FR-0405 | Track project status, workload and overdue tasks

P0 = mandatory only after signed baseline. P1 = optional enhancement subject to separate client approval. Details such as calculation formulas, limits and country-specific rules must be signed off with worked examples.

5. Business Rules, Validation & Data Integrity

- BR-2-01: An employee is linked to exactly one tenant; all references to manager/department/project members must remain in that tenant.
- BR-2-02: A leave request cannot deduct balance until approved; approved balances must reconcile after cancellation/reversal.
- BR-2-03: Concurrent leave approvals must not double-deduct balances; use transactional updates and locking where appropriate.
- BR-2-04: No user can approve their own leave or attendance correction unless client approves an explicit exception.
- BR-2-05: Attendance edits must preserve original event, editor, reason and change history.
- BR-2-06: Cross-company project membership is out of scope unless client explicitly authorizes it.
- BR-2-07: Task dependencies must not form a cycle; archived projects become read-only.

6. Roles, Permissions & Segregation of Duties

Client must approve a role-permission matrix at action level (view, create, edit, approve, export, delete). Default-deny access is enforced by Spring Security and service-layer object authorization. UI restrictions never substitute for API enforcement.

Role	Primary scope (proposed)
HR Admin	Access to assigned module/actions only; final privileges require client-signed matrix.
HR Officer	Access to assigned module/actions only; final privileges require client-signed matrix.
Department Manager	Access to assigned module/actions only; final privileges require client-signed matrix.
Project Manager	Access to assigned module/actions only; final privileges require client-signed matrix.
Team Member	Access to assigned module/actions only; final privileges require client-signed matrix.
Employee	Access to assigned module/actions only; final privileges require client-signed matrix.
Tenant Auditor	Access to assigned module/actions only; final privileges require client-signed matrix.

Control requirements: no unapproved self-approval, separation of preparer/approver where mandated, auditing of privileged changes, and tested cross-tenant isolation for all tenant-owned tables.

7. Business Workflows & User Journeys

- Flow 1: HR admin sets up branch, department and leave policy -> imports/creates staff -> sends invitations -> employee signs in.
- Flow 2: Employee submits leave -> assigned manager approves/rejects -> HR finalizes if required -> entitlement updates and audit event logged.
- Flow 3: Manager assigns shift -> employee check-in/out recorded -> exceptions flagged -> correction approved -> daily summary recalculated.

- Flow 4: Project manager creates sprint -> assigns tasks -> member moves task across statuses -> project dashboard updates.

8. UI Screens & User Experience

Screen group	Minimum proposed interface
Employee & Department Management	List/search/filter with pagination; create/edit/detail forms; permission-aware actions; review, error, empty and audit states
Attendance & Shift Scheduling	List/search/filter with pagination; create/edit/detail forms; permission-aware actions; review, error, empty and audit states
Leave & Holiday Management	List/search/filter with pagination; create/edit/detail forms; permission-aware actions; review, error, empty and audit states
Project, Sprint & Task Management	List/search/filter with pagination; create/edit/detail forms; permission-aware actions; review, error, empty and audit states

Common UX: accessible input labels, confirmation for irreversible actions, reliable error feedback, mobile-responsive layout, keyboard navigation, loading indicators, localization/currency/timezone where applicable. Wireframes require client approval.

9. Data Model & MySQL Entities

Domain module	Proposed core tables (tenant-scoped unless global)
Employee & Department Management	employees, departments, branches, designations, employee_job_history, employee_contacts
Attendance & Shift Scheduling	shift_templates, shift_assignments, attendance_events, attendance_daily, attendance_adjustments, holiday_calendars
Leave & Holiday Management	leave_types, leave_policies, leave_balances, leave_requests, leave_approvals, leave_adjustments
Project, Sprint & Task Management	projects, project_members, sprints, tasks, task_dependencies, task_comments, time_entries

Schema principles: InnoDB foreign keys and appropriate unique constraints; tenant_id on tenant-owned business rows, UTC event timestamps, decimal types for money, effective-date history where necessary, optimistic versioning/locking for concurrent state changes, indexes based on measured queries, Flyway/Liquibase migrations. Do not silently cascade-delete finance, HR or audit evidence.

Client data decisions: retention, import source format, data stewardship and field-level privacy must be confirmed before database design approval. A detailed ERD, dictionary and migration/backfill plan are deliverables, not presumed existing artifacts.

10. REST API Contract - Illustrative Scope

Method	Proposed path	Access
GET/POST	/api/v1/org/employees	Verified tenant role; platform-only where /platform
PATCH	/api/v1/org/employees/{id}	Verified tenant role; platform-only where /platform
GET/POST	/api/v1/org/departments	Verified tenant role; platform-only

		where /platform
GET/POST	/api/v1/org/shifts	Verified tenant role; platform-only where /platform
POST	/api/v1/org/attendance/check-ins	Verified tenant role; platform-only where /platform
POST	/api/v1/org/attendance/adjustments	Verified tenant role; platform-only where /platform
GET/POST	/api/v1/org/leave/requests	Verified tenant role; platform-only where /platform
POST	/api/v1/org/leave/requests/{id}/decisions	Verified tenant role; platform-only where /platform
GET/POST	/api/v1/org/projects	Verified tenant role; platform-only where /platform
GET/POST	/api/v1/org/projects/{id}/tasks	Verified tenant role; platform-only where /platform
PATCH	/api/v1/org/tasks/{id}/status	Verified tenant role; platform-only where /platform

API standards: version /api/v1, DTO validation, consistent error envelope and correlation IDs, OpenAPI documentation, stable pagination/filters, idempotency for critical operations, conflict handling (409), throttling for sensitive endpoints and audit hooks.

11. Non-Functional & Security Requirements

Category	Acceptance baseline / to confirm
Availability	Client to specify hosting/SLA, maintenance window and support hours; no implied uptime warranty.
Performance	Confirm p95 read/write latency and concurrent-user workload using realistic records and tenant distribution.
Security	TLS, authorization on every operation, secure secrets, validated uploads/exports, no password/token logging.
Privacy	Personal and financial data minimization; retention, consent, export and deletion policy signed by client.
Reliability	Backups and restore drill, concurrency testing, idempotent events, transactional state changes.
Accessibility	Keyboard-operable WCAG 2.2 AA target where explicitly agreed; browser matrix for Chrome/Edge/Firefox.
Quality	Unit, integration, API-contract, cross-tenant, regression and E2E tests in CI; security review gates.
Observability	Structured logs, metrics, health checks, alerts and audit trail without sensitive values.

12. In Scope / Explicit Exclusions

Included in Phase 2	Deferred or excluded
Employee & Department Management: agreed requirements	Salary computation, payslips, tax deductions and bank payment files
Attendance & Shift Scheduling: agreed requirements	Lead/customer CRM, invoices and general ledger
Leave & Holiday Management: agreed requirements	Warehouse stock, purchasing and asset depreciation
Project, Sprint & Task Management: agreed requirements	Custom workflow designer; Phase 2 uses configured approval

	chains
Employee & Department Management: agreed requirements	Biometric device, GPS tracking or third-party time clock integrations unless contracted

All third-party services, subscriptions, tax/legal assurance, external integrations and migration volumes require written scope and cost approval. AI, ML and generative AI features are explicitly out of scope for EnterpriseOS.

13. Acceptance Criteria & User Acceptance Testing (UAT)

- UAT-P2-01: HR user can create an employee and assign valid department/manager within the same tenant.
- UAT-P2-02: Employee access is denied to another tenant and private HR fields.
- UAT-P2-03: Overlapping leave or insufficient entitlement is handled according to approved leave policy.
- UAT-P2-04: Simultaneous approvals cannot create a negative leave balance.
- UAT-P2-05: Attendance correction retains original and modified values with an audit record.
- UAT-P2-06: Employee can only view assigned projects/tasks except for explicitly approved shared visibility.
- UAT-P2-07: Task cannot be assigned to an inactive user; cyclic task dependency is rejected.

Client approves UAT evidence with expected result, actual result, test data, screenshot/log reference, tester, date and disposition. Critical/high defects must be fixed or explicitly accepted with documented risk before production release.

14. Development Deliverables & Release Gates

Stage	Deliverables	Exit gate
Discovery	Approved BRD/SRS, policy examples, role-permission matrix, dependencies	Requirements sign-off
Design	Wireframes, architecture notes, ERD/data dictionary, API spec, threat/edge-case review	Design sign-off
Build	React pages, Spring Boot modules, MySQL migrations, tests, audit hooks	Functional demonstration
Verify	Automated CI, security/regression tests, data migration checks, UAT runs	UAT approval
Release	Deployment, rollback/backup procedure, API docs, release notes, handover/training	Go-live authorization

Commercial schedule, milestones and cost remain TBD. They depend on approved scope, policy complexity, third-party licensing, existing-data quality, staff capacity and client review cycles. Changes after sign-off use a documented change request with cost and schedule impact.

15. Assumptions, Risks & Open Decisions

Topic	Risk / resolution
Prior phase dependency	Untested upstream permissions or data structures can invalidate this phase; require regression gate.
Policy ambiguity	Unapproved rule formulas or approval paths cause rework; require signed examples.
Tenant privacy	Cross-tenant and overprivileged exports are release blockers; negative tests required.
Data migration	Legacy fields may be incomplete/duplicated; profile and pilot migration before bulk import.

Performance targets	Capacity and realistic datasets are unknown; benchmark after workload approval.
Compliance & legal	Client owns jurisdictional requirements; engineering does not certify legal compliance.
Scope changes	Use issue/decision log and signed change order before adding new integrations/modules.

16. Client Decisions & Requirements Checklist

☐ Which employee fields are mandatory and which are legally sensitive? Answer/reference: _____

☐ How many branches, departments, employment types and approval levels exist? Answer/reference: _____

☐ How is time recorded: manual, import or biometric integration (integration excluded by default)? Answer/reference: _____

☐ What are leave accrual, carry-forward, holidays and weekend calculation policies? Answer/reference: _____

☐ Are overtime calculations informational or payable? (Payroll deferred to Phase 3.) Answer/reference: _____

☐ Does the client need sprint methodology, timesheets, or just Kanban tasks? Answer/reference: _____

☐ Approved roles/permission matrix attached

☐ Wireframes, report examples and validation rules approved

☐ Hosting/security/retention/deployment requirements agreed

☐ UAT cases and authorized signatories agreed

17. Approval & Sign-off

Signatures indicate approval of the stated Phase 2 requirements baseline and acceptance conditions only. This document does not establish pricing, implementation warranties, or regulatory certifications.

Role	Name / signature	Date
Client Sponsor	_____	_____
Client Product Owner	_____	_____
Technical Delivery Lead	_____	_____

End of Phase 2 requirements document | Version 1.0 | Status: Client review draft